

Linear Regression

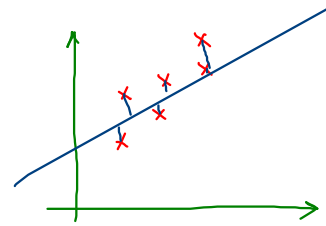
$$y = mx + c$$

m = slope or coefficient
 c = intercept or bias

$$h_{\theta}(x) = \theta_1 x + \theta_0$$

$$W(x) = w^T x + b$$

$$h_{\theta}(x) = \theta_1 x + \theta_0$$



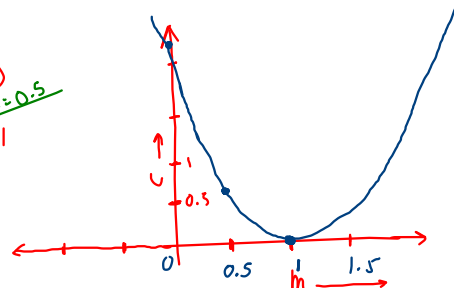
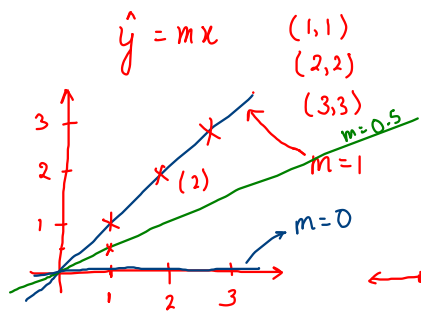
$$\text{Cost function} = \frac{1}{2n} \sum_{i=1}^n (\hat{y}_i - y_i)^2$$

↓

$$\frac{\partial x^2}{\partial x} = 2x$$

$$\hat{y} = mx + c$$

Assumption - Line passes through origin



$m = 0.5$

$$C = \frac{1}{6} [(0.5-1)^2 + (1-2)^2 + (1.5-3)^2]$$

$$\text{Cost} = \frac{1}{6} [0.25 + 1 + 2.25]$$

$$C = \frac{1}{6} [3.50] = 0.59$$

$m = 1$

$$C = \frac{1}{2 \times 3} [(1-1)^2 + (2-2)^2 + (3-3)^2]$$

Cost =

$m = 0$

$$\text{Cost} = \frac{1}{2 \times 3} [(0-1)^2 + (0-2)^2 + (0-3)^2]$$

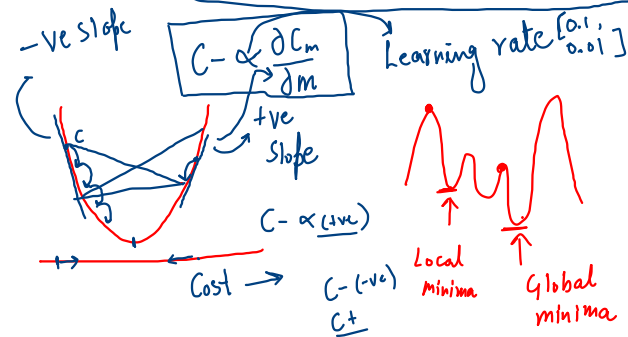
$$C = \frac{1}{6} [1 + 4 + 9] = 2.3$$

m	0	0.5	1
C	2.3	0.59	0

Gradient Descent

$\alpha \uparrow \rightarrow$ Exploding Gradient descent problem

$\alpha \downarrow \rightarrow$ Vanishing Gradient Descent



Regression Model Evaluation Metrics

① Mean Squared Error

Predicted -

Actual - y_i

Dis → Unit change
→ Outliers effect

$$\frac{\sum_{i=1}^m (\hat{y}_i - y_i)^2}{m}$$

Advantage → Penalizes large Error

② Mean Absolute Error

$$MAE = \frac{\sum_{i=1}^m |\hat{y}_i - y_i|}{m}$$

Advantage → No unit change → easy to interpret
→ Robust to outliers

Disadvantage → Do not penalize large errors
→ Direction of error is known

③ Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{MSE}$$

Advantage → No unit change

Dis → Do not penalize large error

④ Coefficient of Determination (R^2 -score)

Range - 0 to 1

$$R^2 = 1 - \frac{SS_{Res}}{SS_{Total}}$$

$$SS_{Res} = \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad SS_{Total} = \sum (y_i - \bar{y})^2$$

where $\hat{y}_i$ is predicted value, y_i is actual, $\bar{y}$ is mean value